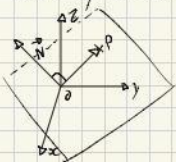


Lecture 6: Square Systems / Equation of planes

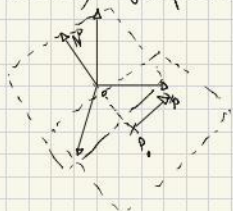
An equation of the form $ax+by+cz=d$ defines a plane

1) Let's say we want to find a plane through the origin with a normal vector equal to $\vec{N} = \langle 1, 5, 10 \rangle$? $\square = \text{plane} \perp \vec{N}$

P is in the plane $\Leftrightarrow \vec{OP} \cdot \vec{N} = 0$
 $\Leftrightarrow x+5y+10z=0$



2) Let's try to find a plane that goes through $P_1(2, 1, 1)$ and $\perp \vec{N} = \langle 1, 5, 10 \rangle$ P is in the plane now $\Leftrightarrow \vec{PP_1} \cdot \vec{N} = 0$



In equation $ax+by+cz=d$

$\langle a, b, c \rangle = \text{normal vector } \vec{N}$

For example: get $\vec{N}$ by cross-product of 2 vectors in the plane

Ex: $\vec{v} = \langle 1, 2, 1 \rangle$; plane $x+y+z=5$

normal vector $\vec{N} = \langle 1, 1, 3 \rangle$, $\vec{v}$ and $\vec{N}$ aren't proportional, therefore, not parallel, so $\vec{v}$ isn't $\perp$ to the plane

$\vec{v}$ parallel to the plane if $\vec{v} \perp \vec{N} \Leftrightarrow \vec{v} \cdot \vec{N} = 0 \Leftrightarrow \langle 1, 2, 1 \rangle \cdot \langle 1, 1, 3 \rangle = 1+2+3=6 \neq 0$
 $\vec{v} \not\parallel$ to the plane

3x3 linear system? example: $\begin{cases} x + z = 1 \\ x + y + z = 2 \\ x + 2y + 3z = 3 \end{cases}$ These two planes intersect in a LINE

The line $\mathcal{P}_1 \cap \mathcal{P}_2$ intersect $\mathcal{P}_3$ in a point = the solution
 Intersection of Plane₁ and Plane₂

Unless, the 3rd plane is parallel to the line where $\mathcal{P}_1$ and $\mathcal{P}_2$ intersect

If line $\mathcal{P}_1 \cap \mathcal{P}_2$ is contained in $\mathcal{P}_3$: infinitely many solutions!!!

If the line $\mathcal{P}_1 \cap \mathcal{P}_2$ is parallel to $\mathcal{P}_3$ (and not in it): no solutions

Matrix A is invertible $\Leftrightarrow \det(A) \neq 0$

Homogeneous Case: id est $AX=0$

e.g. $\begin{cases} x+z=0 \\ x+y=0 \\ x+2y+3z=0 \end{cases}$ There's always an obvious solution: $(0,0,0)$ (origin is a solution because the 3 planes pass through 0).

$\rightarrow$ if $\det(A) \neq 0$, can invert $A \Leftrightarrow AX=0 \Leftrightarrow X=A^{-1} \cdot 0 = 0$

$\rightarrow$ if $\det(A) = 0$: equivalent of saying $\det(\vec{N}_1, \vec{N}_2, \vec{N}_3) = 0 \Leftrightarrow \vec{N}_1, \vec{N}_2, \vec{N}_3$ are coplanar (parallelepiped has volume 0)

The line through 0 and perpendicular to plane of $\vec{N}_1, \vec{N}_2, \vec{N}_3$ is $\parallel$ to all 3 planes and is contained in them $\Rightarrow \infty$ many solutions

Example: $\vec{N}_1 \times \vec{N}_2 = \vec{N}_3$ is $\perp \vec{N}_1, \vec{N}_2$ and also $\vec{N}_3$ so it's a non-trivial solution

General Case: $AX=B$

$\rightarrow$ if $\det(A) \neq 0$ then unique solution $X=A^{-1}B$

$\rightarrow$ if $\det(A) = 0$ then either no sol. or ∞ solutions.

$\det(A)$	$AX=B$	$AX=0$
$\neq 0$	Unique Solution	Unique Sol. $X=0$
$= 0$	0 or ∞ solutions	∞ solutions